

WEEK 1 – TASK 3

ETHICAL HACKING – KALI LINUX LAB SETUP AND LINUX FUNDAMENTALS

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Internship ID: TBS-INT-2026-518

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Task: Week 1 – Task 3 (Ethical Hacking)

1. Introduction

This report documents the practical activity completed during Week 1, Task 3 of the internship, focused on ethical hacking fundamentals. The activity centred on establishing a practical cybersecurity environment using Kali Linux and strengthening basic Linux command-line skills through a guided TryHackMe learning exercise.

2. Objectives

- Set up and access a Kali Linux virtual machine.
- Become familiar with the Kali Linux environment.
- Practice essential Linux terminal commands.
- Understand basic network interface and IP configuration information.
- Update Kali Linux package repositories.
- Develop Linux command-line navigation and file-handling skills.
- Complete the TryHackMe Linux Fundamentals (PT1) room.

3. Lab Environment

The practical activity was carried out using the following environment:

- Host environment: VirtualBox
- Guest operating system: Kali Linux
- Terminal: Kali Linux terminal
- Learning platform: TryHackMe
- Practical module: Linux Fundamentals (PT1)

4. Kali Linux Lab Setup

Kali Linux was used as a virtual machine environment for the practical activity. The Kali Linux VM was started successfully and the terminal was opened. Basic system and network commands were then executed to explore the environment and confirm its readiness for cybersecurity learning and practical exercises.

5. Commands Executed and Their Purpose

Table 1 summarises the Linux commands executed during the practical activity, their purpose, and the observed result.

Command	Purpose	Observed Result
whoami	Identifies the currently authenticated Linux user.	The terminal identified the user as “manuarya”.
ip a	Displays network interfaces, IP addresses, interface states, and related network configuration.	The eth0 interface was active and displayed IPv4 address 10.206.175.48/24.
sudo apt update	Refreshes package repository information so the system can obtain the latest available package metadata.	Repository information was successfully retrieved and the command completed successfully.

6. TryHackMe Linux Fundamentals (PT1)

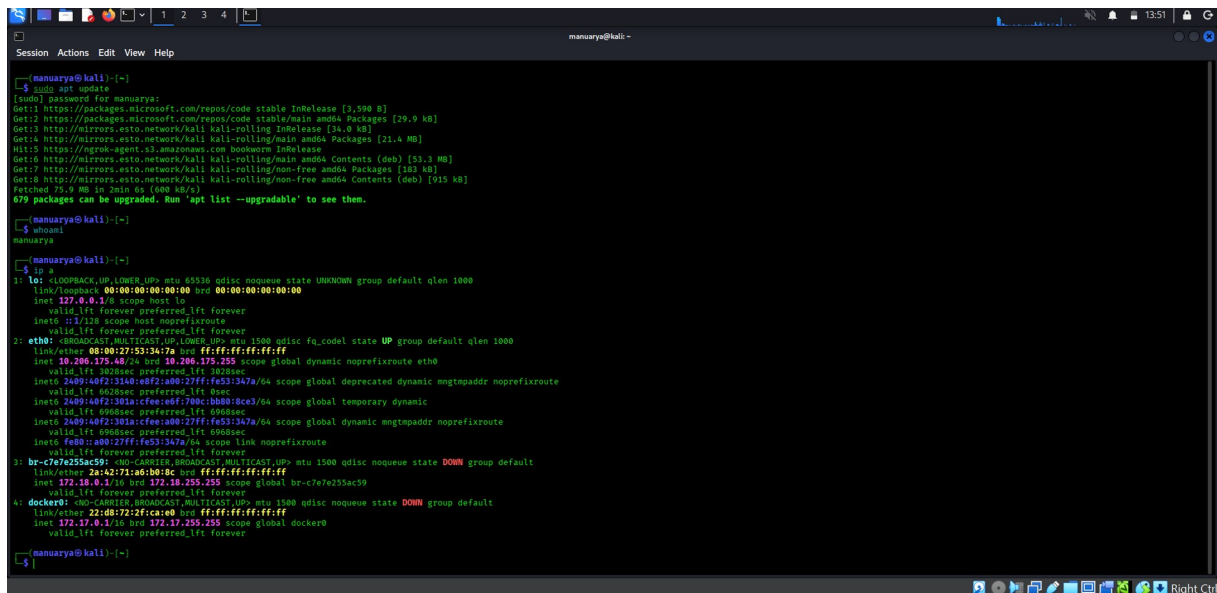
The TryHackMe Linux Fundamentals (PT1) room was completed with 100% completion. The room introduced foundational Linux concepts and provided hands-on practice through an interactive terminal. The following concepts were demonstrated during the room:

- Linux terminal usage
- Directory navigation
- pwd
- ls
- cd
- cat
- grep
- Reading files
- Searching log files
- Identifying information from command-line output

7. Practical Command-Line Exercise

As part of the room, a practical command-line exercise was completed. This involved navigating to the required directory, listing the files present, entering a subdirectory, and reading the contents of passwords.txt using the cat command. The access.log file was then searched using the grep command to locate the required TryHackMe flag, which was successfully identified.

8. Screenshot Evidence



```
manuaria@kali:~$ sudo apt update
[sudo] password for manuaria:
Get:1 https://packages.microsoft.com/repos/code stable InRelease [3,598 B]
Get:2 https://packages.microsoft.com/repos/code stable/main amd64 Packages [29.9 kB]
Get:3 https://mirrors.esto.network/kali kali-rolling InRelease [34.0 kB]
Get:4 https://mirrors.esto.network/kali kali-rolling/main amd64 Packages [31.4 MB]
Hit:5 https://ngrok-agent.s3.amazonaws.com bookworm InRelease
Get:6 https://mirrors.esto.network/kali kali-rolling/main amd64 Contents (deb) [32.3 MB]
Get:7 https://mirrors.esto.network/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:8 https://mirrors.esto.network/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Fetched 75.9 MB in 2min 4s (406 kB/s)
679 packages can be upgraded. Run 'apt list --upgradable' to see them.

manuaria@kali:~$ whoami
manuaria

manuaria@kali:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:33:34:7a/64 brd ff:ff:ff:ff:ff:ff
    inet 10.206.175.40/24 brd 10.206.175.255 scope global dynamic noprefixroute eth0
        valid_lft 3028sec preferred_lft 3028sec
    inet6 2409:40f2:381a:cfee:e6f:700c:b880:8ce3/64 scope global deprecated dynamic mngtmpaddr noprefixroute
        valid_lft 8628sec preferred_lft 8628sec
    inet6 2409:40f2:381a:cfee:a00:27ff:fe53:3a7a/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 8608sec preferred_lft 8608sec
    inet6 fe80::a00:27ff:fe53:3a7a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: br-c7e7e255ac59: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 2a:42:71:a6:b0:8c brd ff:ff:ff:ff:ff:ff
    inet 172.18.0.1/16 brd 172.18.255.255 scope global br-c7e7e255ac59
        valid_lft forever preferred_lft forever
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 22:8b:72:2f:ca:e0 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever

manuaria@kali:~$
```

Figure 1 – Kali Linux Terminal Commands and IP Configuration

Figure 1 demonstrates execution of `sudo apt update`, `whoami`, and `ip a` in the Kali Linux terminal, and shows the Kali Linux network interface information.

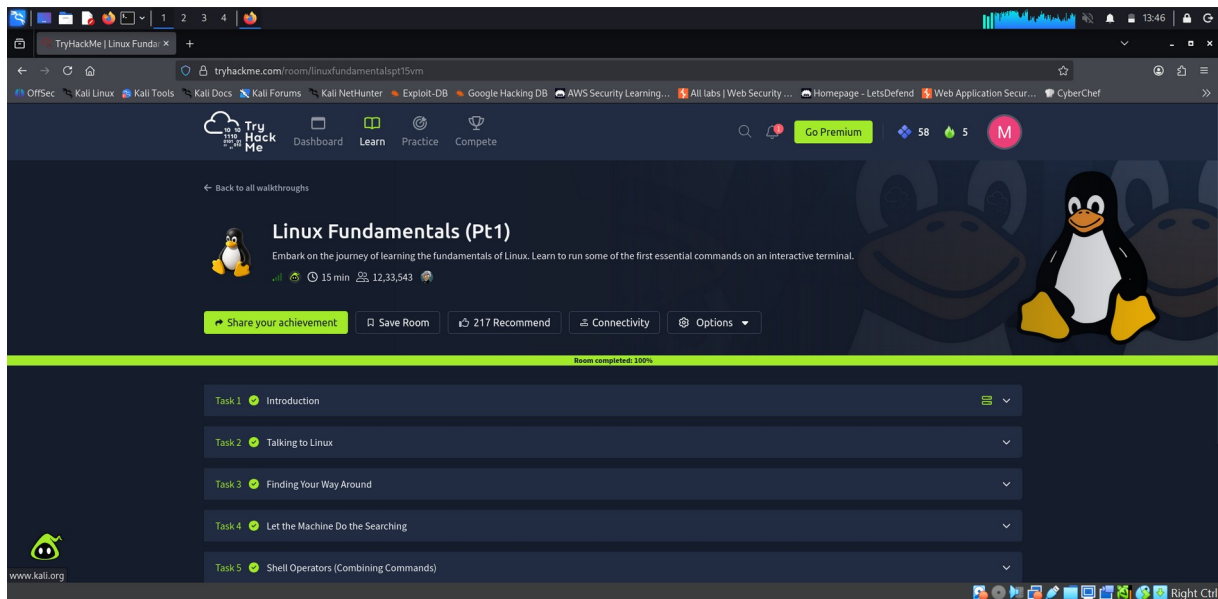


Figure 2 – TryHackMe Linux Fundamentals Completion

Figure 2 provides evidence that the TryHackMe Linux Fundamentals (PT1) room reached 100% completion.

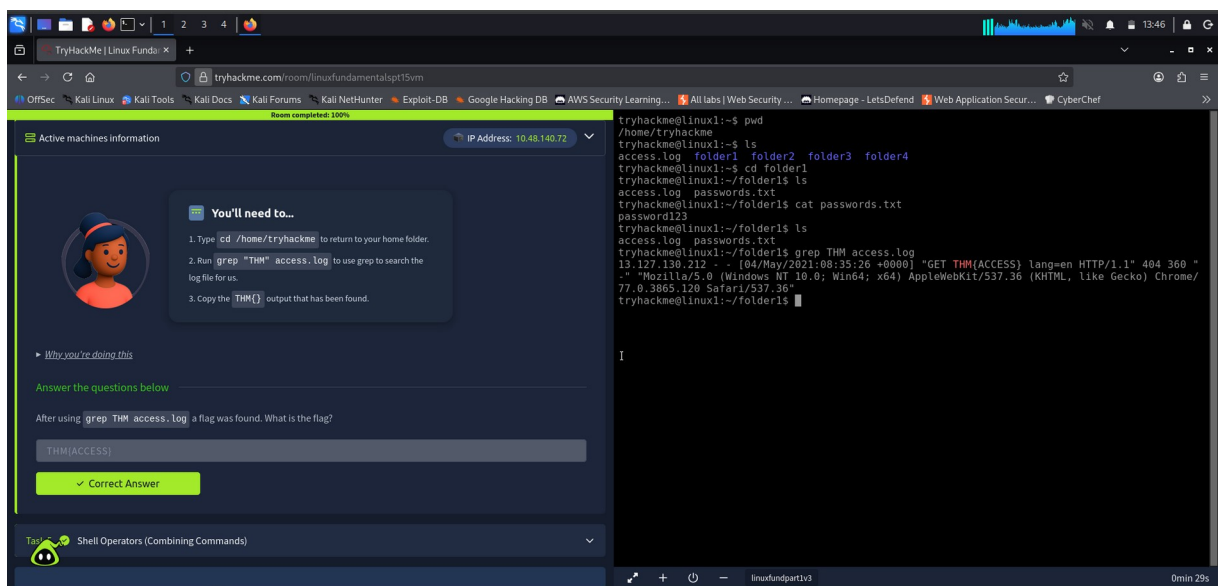


Figure 3 – TryHackMe Linux Fundamentals Practical Exercise

Figure 3 demonstrates practical Linux directory navigation, file inspection, and grep-based log searching, including the terminal activity used to identify the required flag.

9. Skills and Knowledge Gained

- Kali Linux environment familiarity
- Linux command-line usage
- Basic Linux file-system navigation
- File and directory handling
- Log-file searching
- Basic IP and network interface identification

- Package repository management
- Practical cybersecurity lab usage
- TryHackMe-based hands-on learning

10. Practical Learning Outcome

This task provided hands-on experience with Kali Linux and strengthened fundamental Linux skills required for cybersecurity roles. Understanding command-line operations, network configuration, and log searching provides a foundation for later SOC, VAPT, penetration testing, and security monitoring activities. This activity represents an early step in that learning process rather than professional-level expertise.

11. Challenges and Resolution

No major technical issue was encountered during the demonstrated practical activities. The Kali Linux VM started successfully, the terminal commands executed as expected, and the TryHackMe room was completed without any significant obstacles.

12. Conclusion

Week 1, Task 3 was completed successfully. Kali Linux was accessed as a virtualized cybersecurity environment, the required Linux commands (whoami, ip a, and sudo apt update) were executed, and the TryHackMe Linux Fundamentals (PT1) room was completed with 100% completion. This activity strengthened foundational Linux and cybersecurity lab skills that will support future practical tasks in the internship.

13. Declaration

I hereby confirm that the activities documented in this report represent the practical work completed during the Week 1 – Task 3 internship activity. The screenshots included in this report are provided as evidence of the practical work performed.

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